

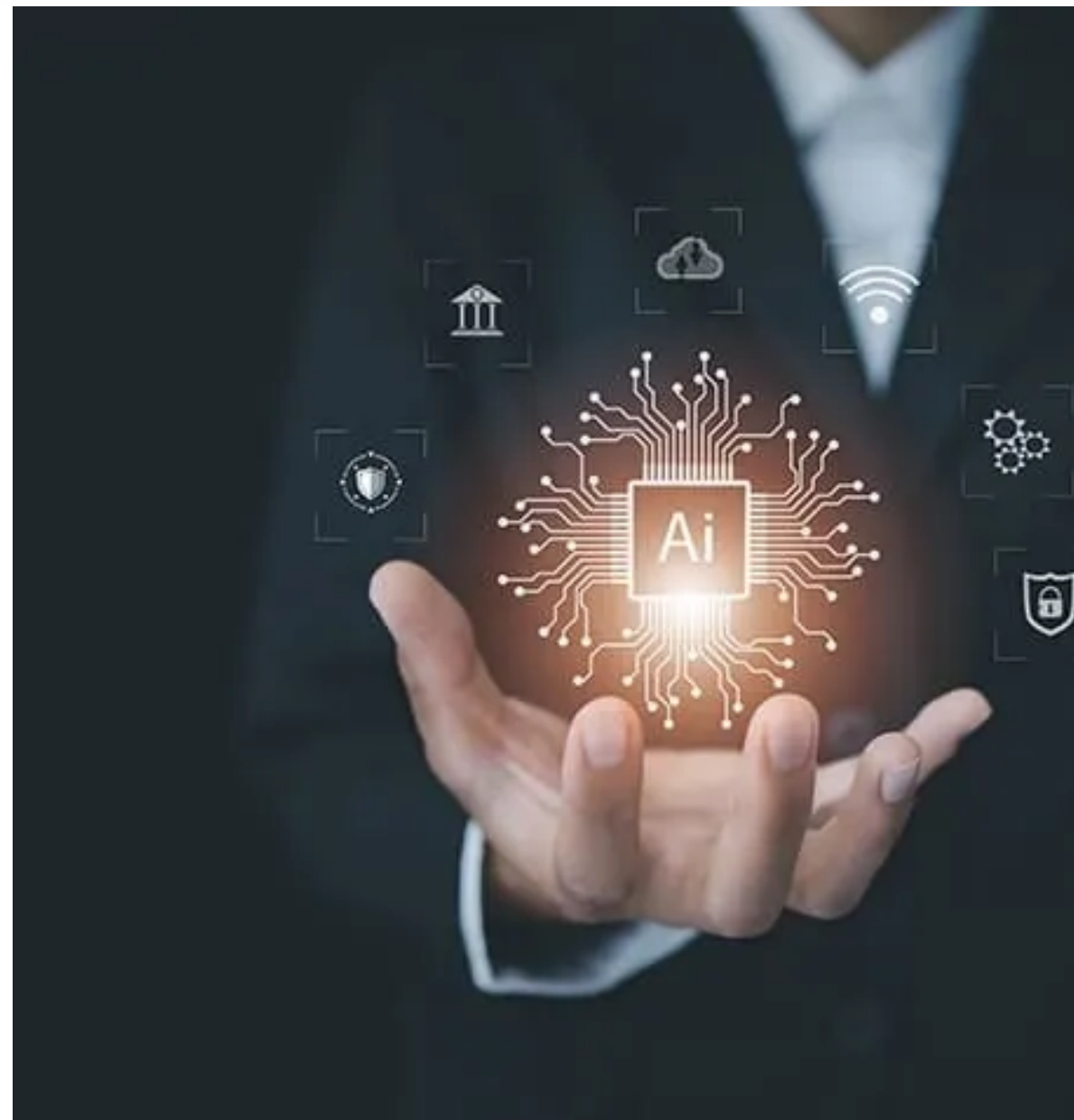
WEEK 4 – LECTURE 2

Artificial Intelligence Foundations

Production Systems & Expert Systems

Rule-Based Reasoning in AI

| CSE276 | Unit II



Quick Recap

In the last lecture, we explored how AI stores information formally.

Knowledge Representation



Semantic Networks



Frames

**Once AI stores knowledge,
how does it actually make decisions?**

Motivation



ATM Machine

Deciding whether to dispense cash to a user.



Doctor

Diagnosing a disease based on symptoms.



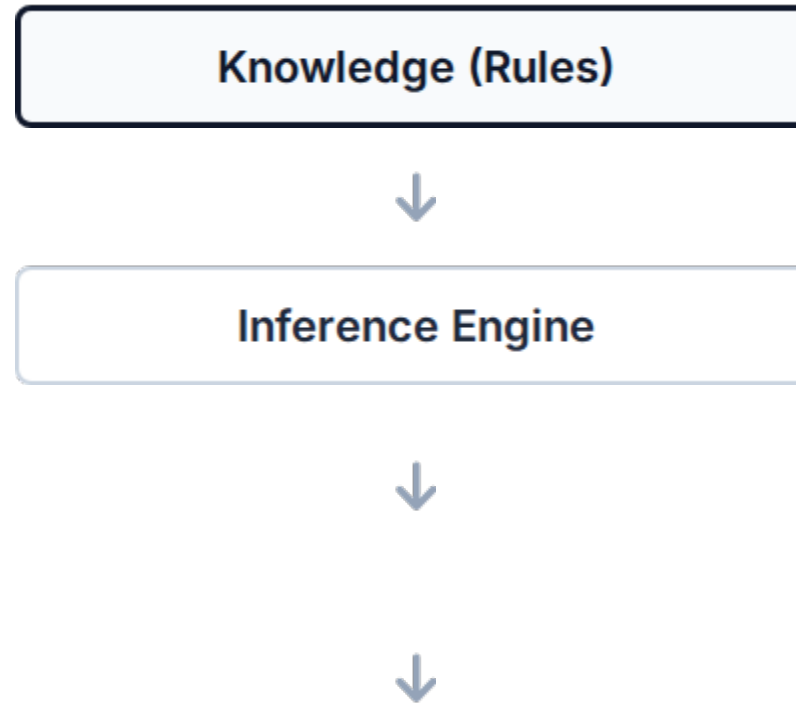
Traffic Signal

Deciding whether vehicles should stop or go.

Can these decisions be represented using rules?

What is a Production System?

A system that applies predefined rules to solve problems and make decisions.



Components of a Production System



Rule Base

The "brain". Stores all the IF-THEN rules provided by experts.



Working Memory

The "notepad". Stores current facts, data, and intermediate results.



Inference Engine

The "processor". Matches rules against the working memory.



Control Strategy

The "manager". Decides what to do if multiple rules match.

Production Rules

The core logic is always written in a simple format:

IF [Condition] THEN [Action]

IF temperature > 38°C
THE trigger fever alert.
N

IF entered_password is correct
THE allow login.
N

IF account_balance >= requested_amount
THE approve ATM withdrawal.
N

Interactive Activity

Convert these real-life scenarios into **IF-THEN** rules:

1. Umbrella usage

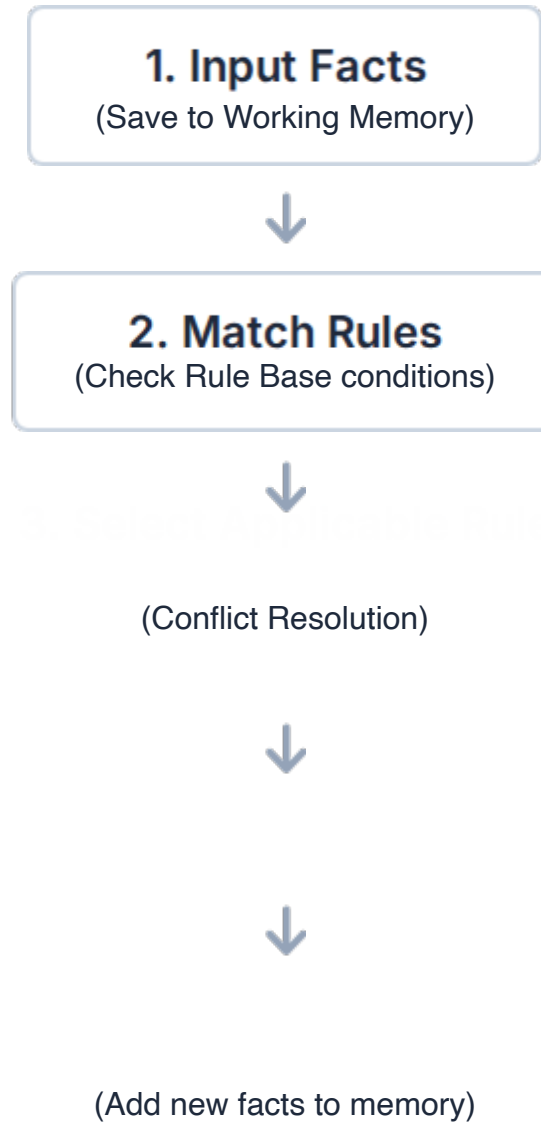
-> IF it is raining THEN take an umbrella.

2. Classroom attendance

-> IF student is present THEN mark attendance as Y.

3. Mobile phone battery warning

How a Production System Works



Rule Execution Example: Medical Diagnosis

Working Memory (Facts)

- ✓ Patient has Fever.
- ✓ Patient has Cough.

Rule Base

IF Fever **AND** Cough
THEN Diagnosis = Flu
AND
IF Diagnosis = Flu
THEN Prescribe Rest and Fluids
END

Conflict Resolution

What happens when multiple rules match the facts simultaneously?



Common Selection Strategies:



Priority

Pick the rule with the highest assigned priority.



Specificity

Pick the most specific rule (most conditions).



First Match

Pick the first rule found in the list.

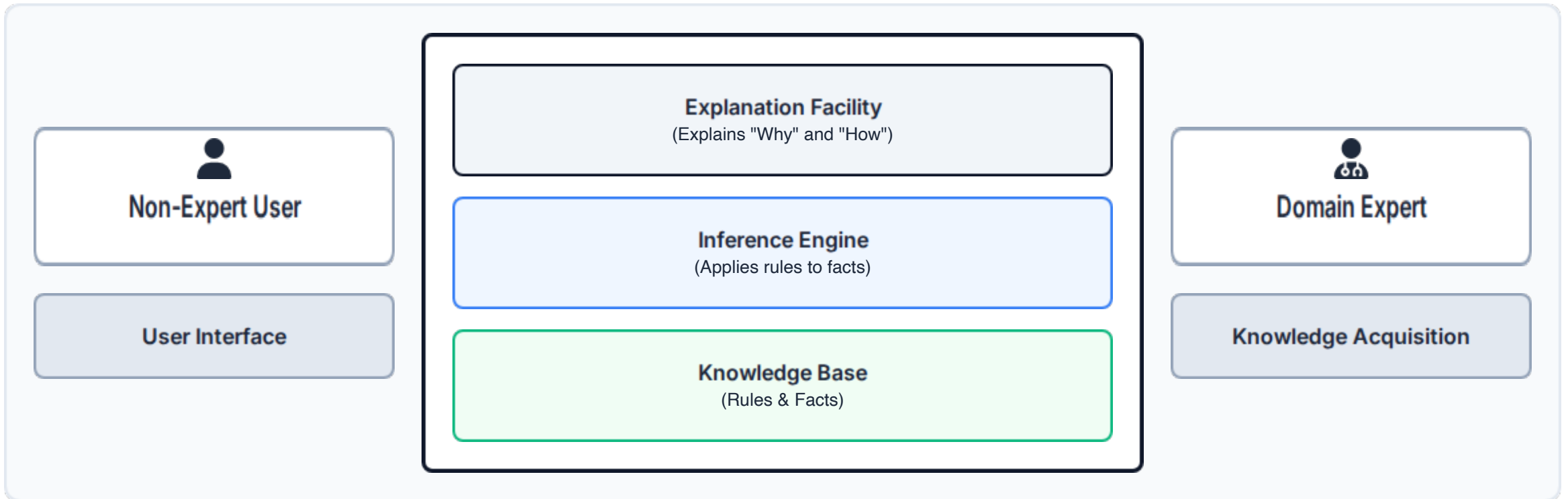


Expert Systems

Can an AI program imitate the knowledge and decision-making ability of a human expert?

Yes! An Expert System is a specific type of production system designed to solve complex problems in a specific domain.

Architecture of an Expert System



Knowledge Base vs Inference Engine

Feature	 Knowledge Base	 Inference Engine
Purpose	The "Memory" of the system	The "Brain" of the system
Stores	Facts and IF-THEN rules	Nothing (Processing only)
Performs	Holds domain expertise	Pattern matching & deduction
Analogy	Medical Textbook	Doctor's reasoning process

How an Expert System Solves a Problem

① User provides symptoms/facts



② Knowledge Base provides rules



③ Inference Engine performs reasoning



Real-World Applications



Medical Diagnosis



Loan Approval



**Tech
Troubleshooting**



**Agriculture
Advisory**



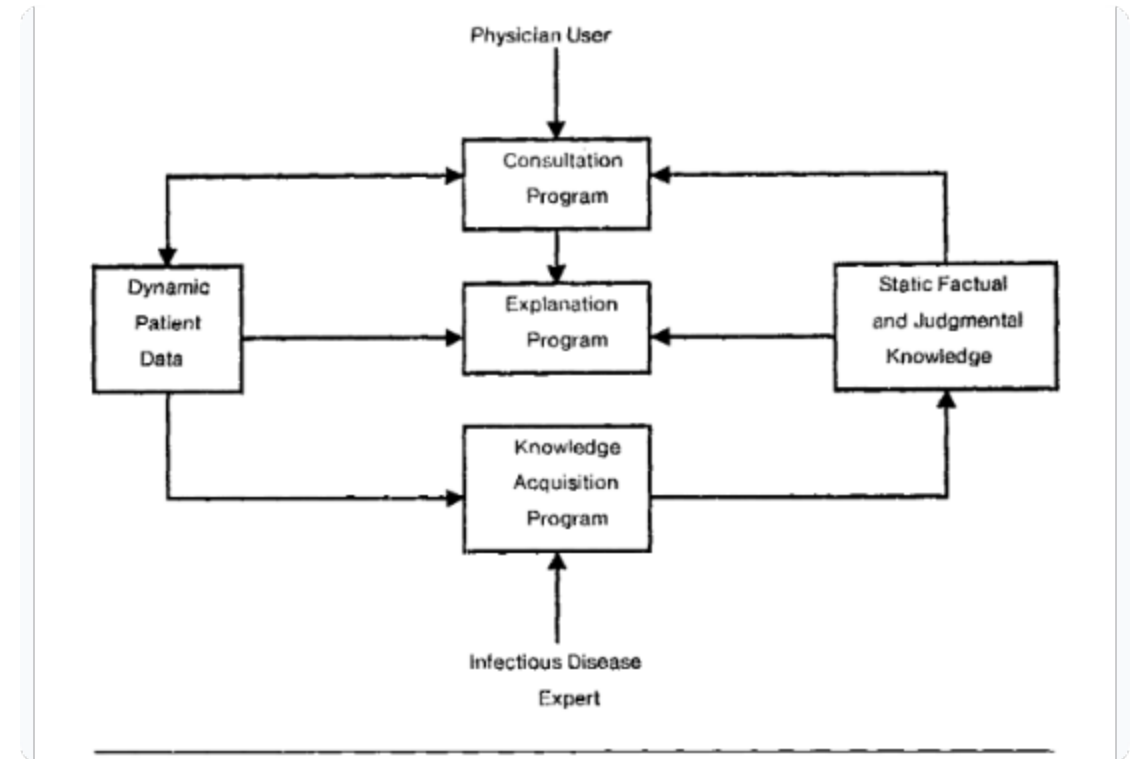
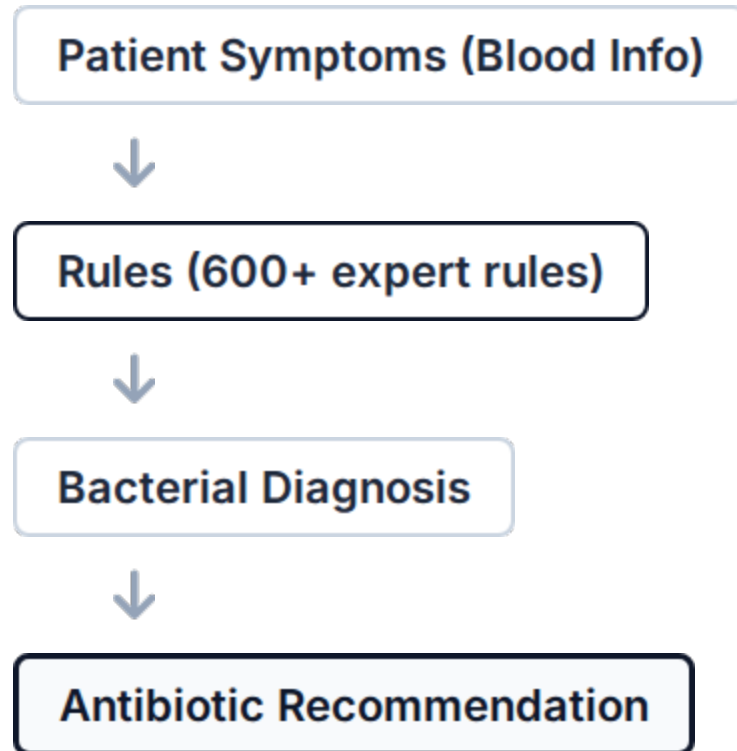
Legal Advisory



Industrial Faults

Case Study: MYCIN (1970s)

One of the earliest and most famous medical expert systems.



MYCIN could perform at the level of human experts.

Advantages & Limitations

Advantages

- ✓ Highly consistent decisions.
- ✓ Fast reasoning.
- ✓ Preserves rare expert knowledge.
- ✓ Available 24/7.

Limitations

- ✗ Difficult/expensive to update.
- ✗ **No learning capability** (Static).
- ✗ Limited to predefined rules.
- ✗ Fails completely on new/unseen problems.



Board Activity

Building a simple Expert System:

Domain: Laptop Troubleshooting

Rule 1: IF won't turn on AND battery dead

→ THEN Charge battery.

Rule 2: IF battery charged AND still won't turn on

→ THEN Check power adapter.

Let's add more rules together!

Common Misconceptions



"Expert systems learn automatically."

False. They only know the rules programmed into them by humans.



"Rule-based AI is outdated and useless."

False. They are still widely used in finance, policy checking, and safety-critical systems.



"Expert systems always make correct decisions."

False. They are only as good as the rules provided. Bad rules = bad decisions.



"Production systems are the same as Machine Learning."

False. Production systems use explicit rules; ML discovers implicit patterns from data.

Key Takeaways



Production systems use
IF-THEN rules.



The Rule Base stores
the knowledge.



The Inference Engine
applies the rules.



Expert systems mimic
domain experts.

Quick Quiz

Which component of an expert system matches facts against the rules to make a decision?

- ☐ A) User Interface
- ☐ B) Knowledge Base
- ☐ C) Inference Engine
- ☐ D) Explanation Facility

Quick Quiz

Which component of an expert system matches facts against the rules to make a decision?

- ☐ A) User Interface
- ☐ B) Knowledge Base
- ☒ **C) Inference Engine**
- ☐ D) Explanation Facility

Preview: Next Lecture

Propositional Logic & First-Order Predicate Logic

How does AI reason mathematically?

- X¹ Logical Statements
- ↻ AND, OR, NOT operations
- A Quantifiers (For All, Exists)

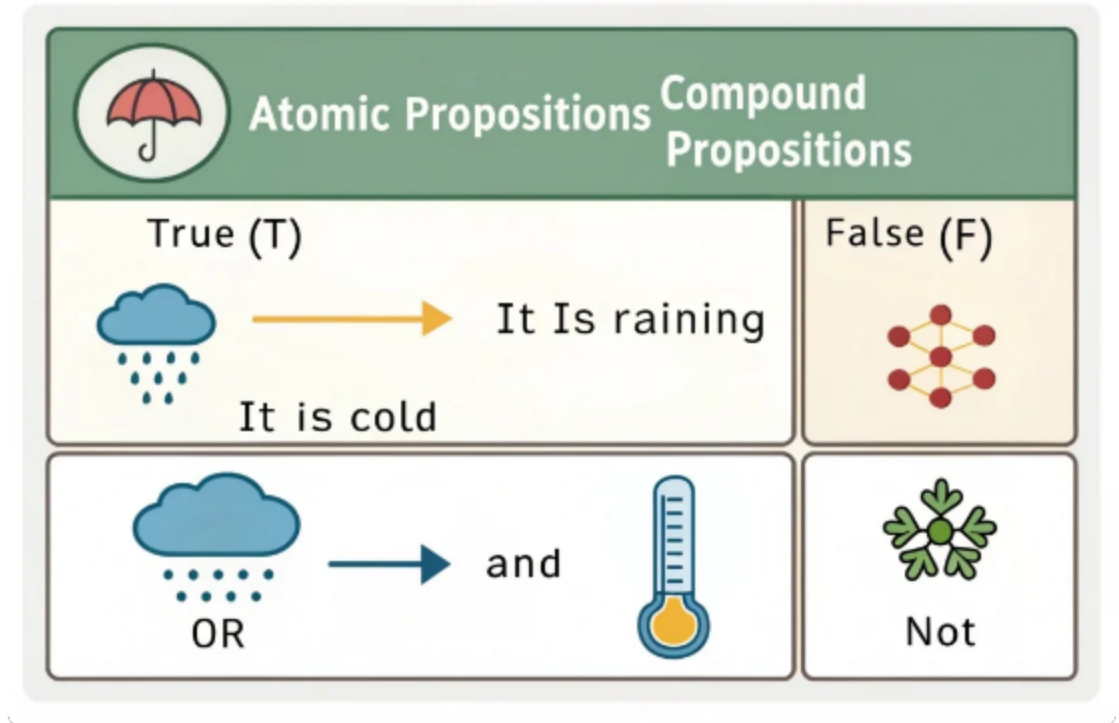


Image Sources



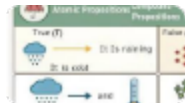
<https://media.geeksforgeeks.org/wp-content/uploads/20260429113421987129/2.webp>

Source: www.geeksforgeeks.org



<https://aithefuture.wordpress.com/wp-content/uploads/2018/05/null.png>

Source: aithefuture.wordpress.com



<https://shoolini.online/blog/wp-content/uploads/2025/06/section-01-propositional-logic-1024x692.webp>

Source: shoolini.online